YEAR 9 PHYSICS, AQA GCSE, UNIT 1, UNIT REVIEW

WORK LIKE A PHYSICIST

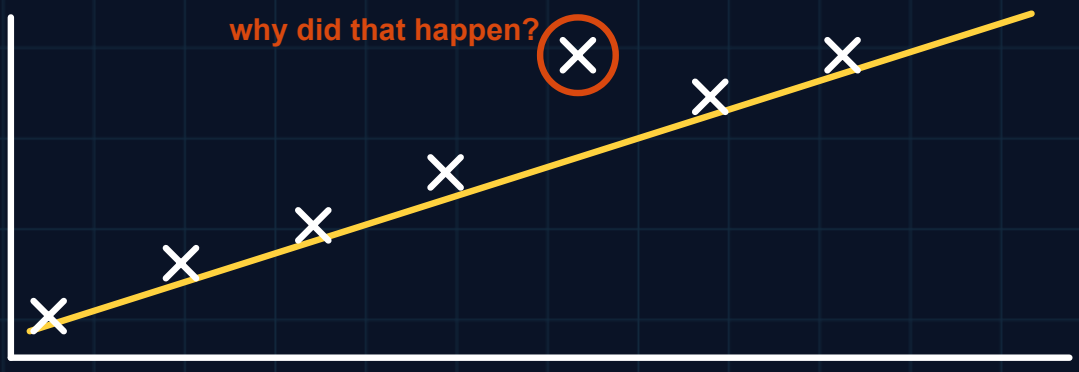
UNIT REVIEW

AFTER LESSONS 7–8

Mixed exam-style questions from the whole unit, a fix-it log and the Physicist's toolkit.

> *"It doesn't matter how beautiful your theory is … If it doesn't agree with experiment, it's wrong."*
>
> RICHARD FEYNMAN, PHYSICIST

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ SCORE / 40 _____

AFTER LESSONS 7–8, AT HOME OR IN CLASS

UNIT REVIEW

Mixed questions from the whole unit, like a real exam. Try each question without looking back. The Physicist's toolkit is on the last page: use it only after you have tried every question. Then mark your answers with your teacher and fill in the "fix it" log.

AQA-STYLE

MIXED EXAM PRACTICE

40 marks, about 45 minutes

1 **Timing a toy car.** A student times how long a toy car takes to roll down a ramp from two different heights.

Ramp height / cm	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
10	1.52	1.48	1.50	1.50
20	1.10	1.14	1.12	1.12

(a) Calculate the mean time for the 20 cm ramp. Write it in the table. **[2 marks]**

(1.10 + 1.14 + 1.12) ÷ 3 = 3.36 ÷ 3 (1)

= 1.12 s (1)

(b) Calculate the uncertainty in the mean time for the 20 cm ramp. Use uncertainty = (largest − smallest) ÷ 2 **[2 marks]**

(1.14 − 1.10) ÷ 2 (1)

= 0.02 s (1)

(c) The stopwatch reads to 0.01 s. Its resolution is **[1 mark]**

0.01 s	0.1 s	1 s

(d) Suggest **one** advantage of using light gates instead of a hand-operated stopwatch. **[1 mark]**

There is no human reaction time when starting and stopping the timer (1).

2 **Keeping it warm.** A student investigates how the **thickness of insulation** around a beaker of hot water affects the **temperature drop** in 10 minutes.

(a) The independent variable is **[1 mark]**

the thickness of insulation	**the temperature drop**	**the starting temperature**

(b) The dependent variable is **[1 mark]**

the thickness of insulation	**the temperature drop**	**the volume of water**

(c) Name **two** control variables. **[2 marks]**

Any two (1 each): volume of water; starting temperature; type of insulation; room temperature; size of beaker.

(d) The thickness of insulation is **categoric** / **continuous** so the results should be shown on a **bar chart** / **line graph** **[1 mark]**

Both needed (1).

3 **Random or systematic?**

(a) Tick **one** box in each row. **[3 marks]**

PROBLEM	RANDOM ERROR	SYSTEMATIC ERROR
A newtonmeter reads 0.5 N with nothing hanging on it.	☐	☑
The person timing reacts a little differently in each trial.	☑	☐
A scale is always read from slightly above, at the same angle.	☐	☑

One mark for each correct row (3).

(b) How should the student correct every reading from the newtonmeter? **[1 mark]**

Subtract 0.5 N from each reading (or zero the newtonmeter first) (1).

4 **Precise or accurate?** Three students measure the length of a rod. Its true length is **50.0 cm**.

Student	Reading 1 / cm	Reading 2 / cm	Reading 3 / cm
A	49.9	50.1	50.0
B	48.2	48.3	48.2
C	47.0	52.5	50.4

(a) Whose readings are precise but **not** accurate? **A** (**B**) **C** **[1 mark]**

(b) Whose readings are both precise and accurate? (**A**) **B** **C** **[1 mark]**

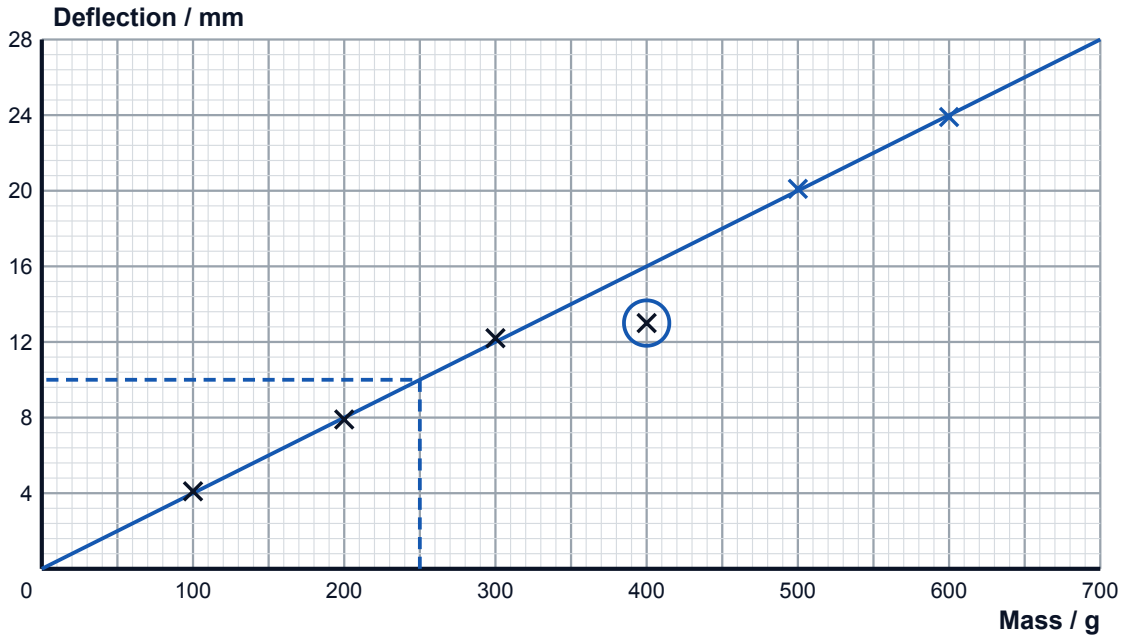
(c) The mean of C's readings is 49.97 cm, very close to the true length. Explain why C's readings are still not precise. **[2 marks]**

The readings are spread out, from 47.0 cm to 52.5 cm, so they are not close to each other (1);

the random errors happen to cancel out in the mean (1).

5 **A sagging beam.** A student hangs masses on the middle of a wooden beam and measures how far it sags (the deflection).

Mass / g	100	200	300	400	500	600
Deflection / mm	4.1	7.9	12.2	13.0	20.1	23.9



(a) Plot the results for 500 g and 600 g on the graph. **[2 marks]**

(500, 20.1) (1); (600, 23.9) (1), each within half a small square.

(b) Draw a ring around the anomalous result. **[1 mark]**

400 g, 13.0 mm (1).

(c) Draw a line of best fit. **[1 mark]**

One ruled straight line through the origin and the other points, ignoring 400 g (1).

(d) Use your graph to find the deflection for a mass of 250 g. Deflection = **[1 mark]**

10 mm

A sagging beam (continued).

(e) What should the student do about the anomalous result? **[1 mark]**

Repeat the 400 g measurement (and leave it out of the line if it is still anomalous) (1).

(f) Write a conclusion. Use the shape of the graph. **[2 marks]**

The deflection is directly proportional to the mass (1)

because the graph is a straight line through the origin (1).

(g) Explain why a line graph is the right choice for these results. **[1 mark]**

Mass, the independent variable, is continuous (1).

6 **Choosing a graph.** Tick **one** box in each row to show how the results should be presented. **[3 marks]**

INVESTIGATION	BAR CHART	LINE GRAPH
How the type of ball affects its bounce height	■	☐
How the temperature of a wire affects its resistance	☐	■
How the colour of a surface affects its temperature rise in sunlight	■	☐

One mark for each correct row (3).

7 **Plan an investigation.** A student wants to find out how the **height** a ball is dropped from affects the **height it bounces**. Describe a method the student could use. Include how to make it a fair test and how to make the results repeatable. **[6 marks]**

Level 3 (5–6): a clear, logical method a stranger could follow, with the variables controlled and repeats used. Level 2 (3–4): a method with some detail, missing a step or a control. Level 1 (1–2): simple statements, not in a logical order. Points: drop the ball from a measured height using a metre rule held vertically; measure the bounce height at eye level (or film it against the rule); change the drop height, at least five values (e.g. 20 to 100 cm); same ball, same surface, release without pushing; repeat each height three times and calculate a mean; check any anomalous result; plot bounce height against drop height.

8 **Whose results can you trust?** Two groups did the bounce investigation. Group A took one reading for each drop height. Group B took three readings and calculated a mean.

(a) Explain why Group B's results are more trustworthy. **[2 marks]**

Repeats show whether the readings are close together / let anomalous results be spotted (1);

the mean reduces the effect of random errors (1).

(b) Group B publish their method and results so that other scientists can check them. This is called **[1 mark]**

a fair test	**peer review**	**a control variable**

REVIEW LATER

FIX-IT LOG

Turn every lost mark into a target.

Q	What went wrong?	The correct physics (or method) is…	Lesson to revisit

My score: _____ / 40. Two things I will practise before the test:

_____

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REFERENCE

THE PHYSICIST'S TOOLKIT

Everything you need for the whole unit, on one page. Fold the corner: you will come back to it.

THE THREE VARIABLES

Independent	the one thing you change
Dependent	the thing you measure
Control	everything you keep the same

Ask: "How does *independent* affect *dependent*?"

TWO KINDS OF DATA

Continuous	any value on a scale: length, time, mass → **line graph**
Categoric	names or groups: material, colour → **bar chart**

Test: is there a value in between?

THE TWO CALCULATIONS

mean = total of the trials ÷ number of trials

uncertainty = (largest − smallest) ÷ 2

Write the result as mean ± uncertainty, e.g. 1.22 s ± 0.03 s.

PRECISE OR ACCURATE?

Precise	repeats close *to each other*
Accurate	close *to the true value*
Resolution	the smallest change an instrument can detect

You can be precise and still wrong.

THE TWO ERRORS

Random	scatters repeats both ways. **Repeat and take a mean.**
Systematic	shifts every reading the same way. **Find the cause; repeats do not help.**

CHOOSING A SCALE

Largest value ÷ number of big squares, then round **up** to 1, 2, 5, 10, 20 or 50 per big square. Equal steps. Use most of the grid.

ANOMALOUS RESULTS

Spot it (far from the trend) → **check** the recording, maths and method → **repeat** if you can. Mistake found: correct it. No cause: keep it, circle it, report it, and leave it out of the best-fit line. Never delete it silently.

CONCLUSION AND EVALUATION

Conclude: "As X increased, Y increased", with two values and units. Straight or curved?

Evaluate: a strength, a weakness, a named error, and an improvement that fixes it. Never just "human error".

GRAPH CHECKLIST

- Title: what against what
- Independent variable on the x-axis
- Both axes labelled, with units: "Ramp height / cm"
- Scale in equal steps that fills the grid
- Means plotted as small, accurate crosses
- One best-fit line: never dot-to-dot
- Bar chart: equal bars, equal gaps, axis from 0
- Anomalous results circled, or "none"

THE RULE THAT DECIDES YOUR GRAPH

Continuous independent variable → line graph. Categoric independent variable → bar chart.

It depends on what you *changed*, not on what you measured.

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